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# A Model-Based Trade-off Analysis for Sizing Integrated Lunar Surface Systems

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**There is an increasing interest in cislunar activities from commercial, public, and military entities, and most depend on lunar surface system capabilities to support or enable their interests. This work develops a library of lunar surface system models, which can be used to size the necessary systems for a lunar architecture based on goals and mission requirements. For the purposes of this study, surface systems are defined as systems whose primary operation occurs on the surface of the Moon, which includes habitats, mobility, in-situ resource utilization, power generation, transportation, and storage. The models are integrated in the Dyreqt framework, which allows resource flows to be created between system. The systems are iteratively sized until the production equals the demand for all resources. A baseline case is sized following the Artemis Planning Manifest for the year 2031. Various parameters, such as the crew size, mission duration, power generation method, and presence of in-situ resource utilization are altered from the baseline to explore potential trade-offs.**

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## I. Nomenclature

| | | |
|---|---|---|
| $\#_{hp}$ | = | Number of heat pipes |
| $\#_{layers}$ | = | Number of MLI layers |
| $\#_{rovers}$ | = | Number of extraction rovers |
| $\alpha$ | = | Absorptivity of aluminum |
| $\Delta G$ | = | Gibbs free energy |
| $\epsilon$ | = | Emissivity of aluminum |
| $\eta_{bat}$ | = | Battery efficiency |
| $\eta_{bol}$ | = | Beginning-of-life efficiency |
| $\eta_{deg}$ | = | Yearly degradation |
| $\eta_e$ | = | Electrolyzer efficiency |
| $\eta_{eol}$ | = | End-of-life efficiency |
| $\eta_{fc}$ | = | RFC efficiency |
| $\eta_{inherent}$ | = | Inherent degradation |
| $\eta_w$ | = | Wire efficiency |
| $\theta$ | = | Angle of incidence |
| $\nu$ | = | Poisson ratio |
| $\rho_0$ | = | Base resistivity |
| $\rho_{Al}$ | = | Density of aluminum |
| $\rho_d$ | = | Dielectric resistivity |
| $\rho_{emi}$ | = | EMI shielding density |
| $\rho_w$ | = | Wire resistivity |
| $\rho'_{BetaCloth}$ | = | Areal density of beta cloth |
| $\rho'_c$ | = | Areal density of the cylindrical inflatable section |
| $\rho'_d$ | = | Areal density of the dome inflatable section |
| $\rho'_{MLI}$ | = | Areal density of MLI |
| $\rho'_{MMOD}$ | = | Areal density of MMOD shielding |

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$\rho'_{PV}$ = Areal density of PV array
$\sigma_{sb}$ = Stefan-Boltzmann constant
$\sigma_y$ = Yield stress of aluminum
$A$ = Cross-sectional wire area
$A_i$ = Area of the inflatable part of the dome section
$A_{PV}$ = PV array area
$D_d$ = Discharge depth
$e_{bat}$ = Specific energy of the battery
$E_{req}$ = Required energy storage
$eol$ = End-of-life value
$F$ = View factor
$H_c$ = Height of the cylindrical section
$H_d$ = Height of the dome section
$I$ = Current
$k_d$ = Dielectric conductivity
$k_w$ = Wire conductivity
$L$ = Length of the power cable
$L_{hp}$ = Length of the heat pipe
$L_{store}$ = Storage length of the spool
$M$ = Bending moment
$m_c$ = Mass of the cylindrical section
$m_{cable}$ = Power cable mass
$m_d$ = Mass of the dome section
$m_{H2}$ = Hydrogen mass
$m_{hp}$ = Mass of the heat pipes
$m_{int}$ = Mass of the lander integration structure
$m_{legs}$ = Mass of the habitat ground support structure
$m_{non-opt}$ = Non-optimal mass of the habitat
$m_{O2}$ = Oxygen mass
$m_{out}$ = Outfitting mass
$m_{PV}$ = Mass of the PV array
$m_{rad}$ = Mass of the radiator
$m_{ring}$ = Dome-to-cylinder interface ring mass
$m_{ss}$ = Mass of secondary structures
$m_{tot}$ = Total mass of the habitat
$\dot{m}_{h2o}$ = Mass flow rate of water
$\dot{m}_{o2}$ = Mass flow rate of oxygen
$\dot{m}_r$ = Mass flow rate of regolith
$p$ = Internal pressure
$P$ = Power load
$P_e$ = Power used by the electrolyzer
$p_{eol}$ = End-of-life power generation per square meter
$P_{lost}$ = Power lost
$P_{req}$ = Power generation required
$P_{rover}$ = Power used to charge the rovers
$q_{gen}$ = Heat generation per cubic meter
$q_{in}$ = Incident radiation
$q_{rad}$ = Heat generation from radiation
$R$ = Resistance
$R_c$ = Radius of the habitat
$r_d$ = Outer radius of the dielectric
$r_i$ = Radius of the inner conductor
$r_o$ = Outer radius of the outer conductor
$R_{reg}$ = Resistance of the regolith

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| | | |
|---|---|---|
| $SF$ | = | Safety factor |
| $t$ | = | Operational lifetime |
| $t_c$ | = | Thickness of the aluminum cylinder |
| $t_d$ | = | Thickness of the dielectric |
| $T_d$ | = | Wire temperature at $r_d$ |
| $t_{dark}$ | = | Time in the darkness |
| $T_i$ | = | Wire temperature at $r_i$ |
| $T_o$ | = | Wire temperature at $r_o$ |
| $t_p$ | = | Plate thickness |
| $T_s$ | = | Temperature of the surface regolith |
| $T_{space}$ | = | Temperature of space |
| $t_{sun}$ | = | Time in the Sun |
| $T_w$ | = | Wire temperature |
| $V$ | = | Voltage |
| $V_b$ | = | Breakdown voltage of the dielectric |

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# II. Introduction

With many agencies such as NASA, ESA, and JAXA working towards landing humans on Mars, there has been a resurgence in lunar interest due to the current Moon-to-Mars architecture. This relies on the Moon as a proving grounds, to test strategies and capabilities before proceeding to send humans to the Martian surface. The Moon is the nearest celestial body, making it the ideal place to test new technologies. Before humans can land and survive on the Martian surface, there are many vital technologies that must be matured to verify that they operate as expected in an extraterrestrial environment[1]. The lunar interests of the government agencies is mirrored by the commercial sector, as the creation of many of the lunar systems will be contracted out to aerospace companies[2]. Before the systems can be built, it is necessary to have requirements that ensure the systems will effectively operate together. This includes sizing them such that the resource production and demand are balanced.

## A. Lunar Surface Systems

Lunar surface systems are systems whose functionality and primary operation occur on the surface of the Moon. This encompasses a wide variety of systems, like exploratory/scientific rovers, pressurized rovers (mobile habitats), unpressurized rovers, power generation systems, and in-situ resource utilization (ISRU) systems. Permanent habitats are also surface systems, but not temporary habitats like lunar landers (HLS)[3]. A lunar base refers to a specific combination of surface systems, and a lunar architecture is a lunar base with the layout and connections between systems defined, an example of which is shown in Fig. 1.

The systems that are included in Fig. 1 are the same systems that are being modeled in the baseline case. There is a surface habitat that will allow the astronauts to live on the surface for weeks without resupply, a pressurized rover that can provide life support for a few days to enable longer exploration missions, an unpressurized rover for short trips near the base, batteries to store energy, and a photovoltaics system to generate power.

A lunar base is designed to satisfy stakeholder needs that are defined by a set of requirements, and different combinations of surface systems can fulfill the same set of requirements. An example requirement is: the lunar base must generate 10 kW of power; a photovoltaic array or fission reactor can both achieve this. Therefore there are trade-offs that exist, and trade studies must be performed to design the ideal base to meet the stakeholder needs. The purpose of this research is to investigate the trade-offs between different technological choices and explore the sensitivities of mission parameters when sizing different lunar architectures. There are complex sizing dependencies between systems, so the effects of changing mission parameters will depend on the architecture being investigated.

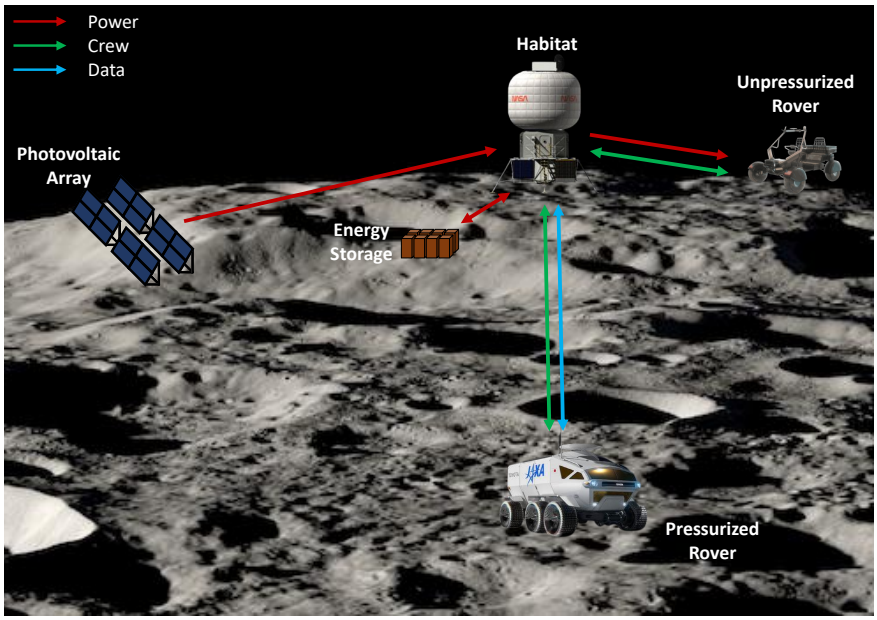



**Fig. 1 An example lunar architecture[4–7].**

### B. Background

Significant research exists on the design of lunar surface systems. Lunar habitats have been researched since the 1960s, many focusing on only one disciplinary aspect. There have been studies investigating the effects of habitat layout and emplacement[8], structural design[9], Environmental Control and Life Support System (ECLSS) trade-offs[10], and thermal control system design[11, 12]. These studies go in depth on small parts of a lunar base, but they do not consider the larger habitat system or the interactions with other systems, missing the important relationships being investigated in this study.

Reference [13] performs a sizing analysis of a lunar base power generation and energy storage system with and without oxygen produced by ISRU. Photovoltaics or a fission reactor are used to generate power, and batteries or fuel cells are used to store power. However, the base is located at 30° North, so the day-night cycle and temperature assumptions are very different from the lunar South Pole. Only the power subsystem is being sized, so other subsystems like the ECLSS do not change when ISRU is added. Because the other subsystems are fixed, this does not capture the sizing interactions that are of interest in this paper. Another power sizing paper examines a power system for an early lunar base near the South Pole by the Shackleton Crater rim. It finds the necessary power generation value to supply ISRU, science, rover charging, and RFC loads, with losses from transmission and inefficiencies. There are three options for power storage: RFCs, batteries, and flywheels. For transmission, both AC and DC transmission is considered, as well as high and low voltages. This study is focused mainly on power transmission and storage trades, and does not consider power generation trades, let alone any trades relating to the surface habitat or other systems[14]. There are other studies that focus solely on power generation trades, but again these do not model changes for other systems[15, 16].

An ISRU study investigated the sizing of the ECLSS when production from ISRU was included. It also compared different landing sites at the lunar South Pole, and how it affected both systems. The power system was out of scope, and a fixed ECLSS architecture was used. The focus on the impact of ISRU on a specific ECLSS architecture covers only a small fraction of the total cases modeled in this paper[17].

The reviewed studies have a narrower focus and do not size the entire lunar architecture. Furthermore, all of the studies examined above either size a single design or a few case studies. None of them perform parameter sweeps to determine the system sensitivities, or look at every possible architecture to visualize the trade-offs. The work presented in this paper addresses these shortcomings.

### C. Motivation and Research Objectives

There is a need for a preliminary sizing tool that sizes the systems for a lunar architecture in an integrated way, rather than individually. This integrated approach has many advantages: it ensures no systems are over- or undersized



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and metrics for the entire architecture can be calculated. When individually sizing systems, any change to one system causes imbalance with the other systems. One way to account for this is with margins, oversizing the systems to be able to handle more demand than expected. This overestimates the mass and volume of the systems, both of which are some of the most common limiting factors for space systems. Another approach is to resize the other systems, which will require multiple reruns to converge all systems. By integrating the systems and their resource demands, they can all be solved simultaneously. The results of this approach can help inform mission feasibility, and requirements, as well as provide recommendations for technologies that future research and development efforts should focus on.

There are many different possible lunar architectures, and they will likely be built up and evolve over time. The first systems will likely be rovers and temporary habitats, and over time power systems, permanent habitats, and ISRU capabilities will be added to the architecture. Eventually there could be multiple bases, each with their own surface systems and interactions. An integrated sizing tool must be modular so that systems can easily be added or removed, enabling the model to evolve alongside the architecture. It also must handle multiple instantiations of the same system, to account for duplicates, or to size the same system for different functions. The modularity also enables trade studies, allowing deviations from the baseline to be quickly modeled for comparison. The surface system models must be parametric in order to support mission parameter variations when sizing; being able to change variables like the crew size is essential to capture different/evolving architectures. A modular and parametric sizing tool that integrates lunar surface systems can be used to run trade studies and sensitivity analyses of a lunar architecture, revealing trends that cannot be found with current methods.

An integrated systems analysis tool was created to size all surface elements of a lunar architecture and enable trade studies. To summarize the main research objective for this work:

- Create sizing models for lunar surface systems of interest
- Integrate the models in a parametric solving environment
- Explore trade-offs between different lunar surface system architectures
- Investigate lunar surface system sensitivities to mission parameters

This work supports the Georgia Tech Cislunar Architecture Study, which aims to predict future lunar architectures that achieve certain objectives while operating under a budget. Instead of considering only a single entity, the study is addressing the multi-stakeholder problem and integrating the interests of science, commerce, and security. An interactive tabletop exercise for planning programs and missions will be run in real-time, using surrogate models to size the architecture and predict costs. The creation of surrogates necessitates the sizing models to be parametric, in order to easily size architectures across the design space[18].

## III. Approach

First the library of lunar surface system models was created. The models take high-level inputs from the mission, like the crew size and technology choices, as well as resource demands from other systems. The models output a sized system, calculating the mass of the system and resource loads that other systems must handle. A combination of regressions from literature and physics-based relationships are used to create the models; most are broken down into submodels to separate the different disciplines when sizing.

Once the models have been created, they are implemented in the Dyreqt 2 framework. Dyreqt is an object-oriented space systems sizing and synthesis extension of OpenMDAO in Python, which is currently being developed by the Advanced Concepts Office at NASA MSFC. The lunar base is modeled as an element, and the various systems are modeled as subelements or groups of subelements. For example, the habitat is a group of subelements, comprised of individual subelements like the structure, thermal control loops, radiator, science payload, etc., and further subelement groups like the life support system. Dyreqt allows the user to define networks with the subelements as nodes and resource flows as edges. Example resource flows include power, heat, water, and oxygen. Each subelement in a resource graph interacts with the resource in some way, by either contributing or removing an amount of the resource. Dyreqt converges the size of all subelements such that the resource flows are balanced[19]. The sizing methodology using Dyreqt is shown in Fig. 2.

A baseline case was run to use as a starting point for comparisons. The Artemis Planning Manifest was used to create the baseline case; it outlines a plan to have a lunar base with the necessary power generation, an unpressurized lunar rover, and a pressurized lunar rover on the Moon by 2031 [3]. Finally, the baseline architecture was altered, changing parameters such as the crew size, power generation type, and the addition of ISRU to see how the lunar surface systems change relative to the baseline values. The baseline case and variations are described in detail later on in Section III.B.



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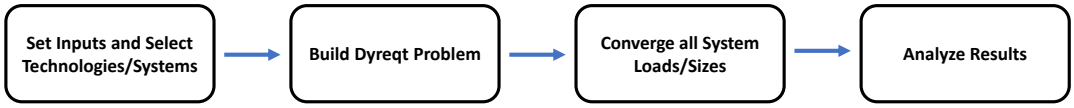



**Fig. 2 Lunar surface systems sizing methodology.**

### A. Model Architecture

The high-level systems being modeled include the habitat, pressurized and unpressurized lunar rovers (PR and UPR), power generation, power distribution, energy storage, and in-situ resource utilization. The habitat is broken down into subsystems, which includes models for the environmental control and life support systems, thermal control, and structures. When sizing, the thermal control is further broken down into the internal thermal control system (ITCS), external thermal control system (ETCS), and radiator. Likewise, other systems and subsystems are also broken down by function or discipline when sizing, like the life support, pressurized rover, unpressurized rover, power distribution, and ISRU. A condensed n-squared diagram, with only the habitat partially expanded, is shown in Fig. 3.

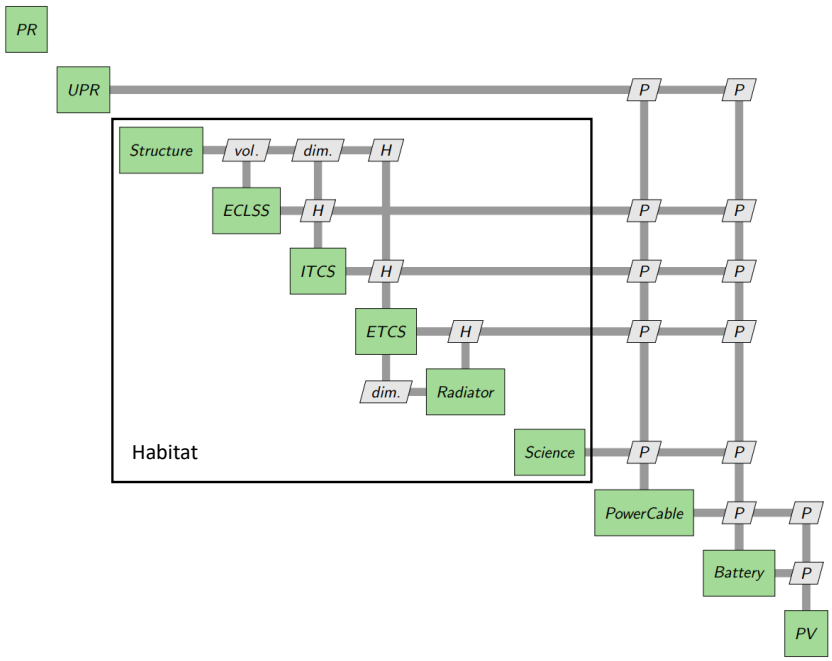



**Fig. 3 Power (P), heat (H), volume (vol.), and dimension (dim.) information flows of higher level systems.**

The pressurized rover is not connected to other systems in the diagram, because it is designed to be independent to perform long missions away from the habitat. The pressurized rover has its own solar array so it is assumed to not take power from the grid, and also assumed to have no life support resource connections with the habitat. The unpressurized rover recharges its batteries by connecting to the grid, because it does not have a photovoltaic array. The other systems are connected through heat, power, or other resources. The connections are made at the lowest level for all systems when sizing. An example base layout with the associated power and thermal layers is shown in Fig. 4.

Just like in Fig. 3, the pressurized rover does not connect to the power grid. The other systems require power from the solar array or batteries, and the only two-way connection is between the habitat and the batteries. All of the surface systems have thermal control except the solar arrays, which do not need cooling. Besides the habitat and batteries, all of the thermal control systems are independent. The habitat has multiple internal and external subsystems that require cooling, and the batteries are assumed to be part of the external fluid loop. The pressurized rover is essentially a mobile habitat, so it also has internal and external fluid loops to cool different subsystems.



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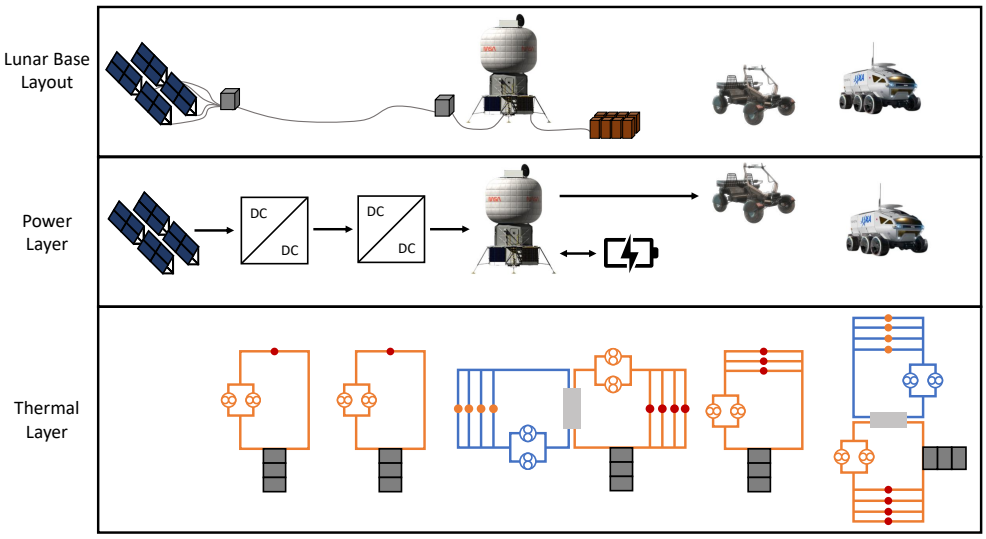



**Fig. 4 Power and thermal connections for a notional base layout[5–7].**

*1. Environmental Control and Life Support System*

The lunar surface habitat is a vital part of a manned mission architecture, as it allows the crew to survive for a long duration on the surface. One part of the habitat is the ECLSS, which is composed of many subsystems that contribute to keeping the conditions inside the habitat livable. This includes revitalizing and cleaning the air, managing waste, storing and recycling water, fire suppression, providing food, and maintaining the pressure. The Advanced Life Support Sizing Analysis Tool (ALSSAT) was created by NASA for the preliminary sizing of life support systems, using technologies employed by the International Space Station. The tool was created in Excel; the sizing relationships were re-written in Python so the tool could be used in an Dyreqt subelement.

ALSSAT requires inputs from the mission, like the size of the crew, number of extra-vehicular activities, and the total mission duration, as well as inputs from the habitat, like the number of modules and the total pressurized volume. The tool has multiple technology options for the water management system, oxygen generation system, and the carbon dioxide removal and reduction systems, which are listed in Table 1. Some of the technologies are regenerative, meaning they recycle waste products, allowing resources to be recovered for reuse.

**Table 1 Life support system alternative technologies.**

| System Function | Alternative 1 | Alternative 2 | Alternative 3 |
|---|---|---|---|
| $CO_2$ Removal | 4-Bed Molecular Sieve | CAMRAS | LiOH Canisters |
| $CO_2$ Reduction | None | Sabatier | Bosch |
| Oxygen Management | Oxygen Storage | Solid Polymer Electrolysis (SPE) | High Pressure SPE |
| Water Management | Water Storage | Water Recovery | |

ALSSAT has many built in assumptions, like module leak rates, crew member consumption and production rates, and system efficiencies that, when combined with the other inputs and the technology choices, result in resource flows. ALSSAT uses a resource mass balance algorithm to ensure that the net change in every resource is zero. For example, the oxygen consumed by the crew and lost through leaks is produced at exactly the same rate by the oxygen generation assembly, so the oxygen level is unchanged. ALSSAT is a static sizing tool, so all consumption and production rates are held constant throughout the mission. Each individual component is sized by the resource flow it must account for. The decomposition of ALSSAT is shown in Fig. 5 [20].

The parts are scaled by the resource demand that the component must meet. Each part has a unique sizing equation with the resource demand as an input, which results in a mass, volume, power load, and a heat load. The properties of a component are the sum of its constituent parts, and following this bottom-up approach the properties of the functions,



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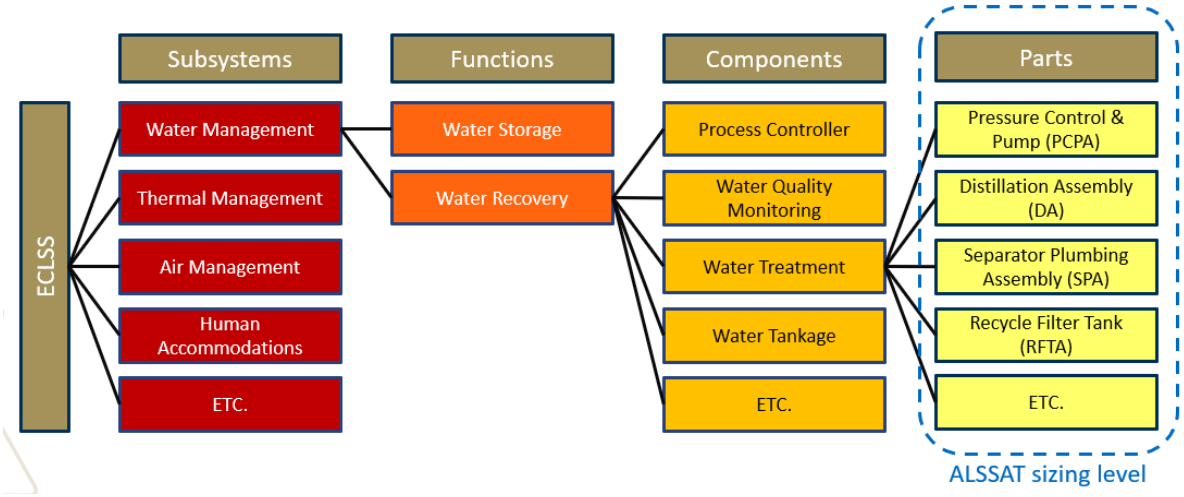



**Fig. 5 ALSSAT example subsystem breakdown and sizing level [21].**

subsystems and total ECLSS can be found[20].

*2. Thermal Control*

The lunar surface temperature varies from 90 to 390 K, so the surface systems must have a robust thermal control subsystem to regulate their temperature. The lunar regolith has a conductivity of only 0.04 W/m$^2$-k, making it an excellent insulator, so the best way to dissipate heat is through radiation[11]. Solar radiation, lunar albedo, and earthshine are the main sources of external heat generation, with solar radiation being the greatest. Passive thermal controls like multi-layer insulation (MLI) and highly reflective paints are used to minimize the heat gain from external sources. There are also sources of internal heat generation; the electronics generate heat due to inefficiencies and for a habitat, the crew generate heat through metabolism.

A pumped fluid loop system is used for applications that require large amounts of heat dissipation. An example is the active thermal control system (ATCS) for the habitat. It is comprised of an internal pumped fluid loop with water as the working fluid and an external loop with ammonia. The internal loop operates at a lower temperature and removes heat from the cooler internal systems, like the ECLSS. The external loop has a higher operating temperature and removes heat from any external systems, like batteries, as well as any internal systems that operate at a higher temperature. The external loop also removes heat from the internal loop via a heat exchanger. The external loop gets rid of the excess heat by flowing through the radiators, which radiate the heat to the surroundings. Cold plates and heat exchanges are sized to transfer the heat loads from the systems to the loops. A diagram of the thermal control system for the habitat is shown in Fig. 6. The pumped fluid loops and radiators are sized using the models described in Reference [22].

A heat pipe thermal control system is used for applications that do not generate much heat. The working fluid is circulated with a wick instead of a pump, so it does not consume any power. Depending on the amount of heat removal required, multiple heat pipe systems can be used. The number of heat pipes can be calculated using Eq. 1 [23].

$$\#_{hp} = ceil(\dot{Q}/240) \tag{1}$$

The heat pipe system mass includes the mass of the pipes, the cold plates used to transfer heat, and the working fluid. The total mass can be calculated using Eq. 2 [23].

$$m_{hp} = \#_{hp}[24 + 2.94 \times 10^{-4} \cdot 240 L_{hp}^2 + (24 + m_{rad})/20] \tag{2}$$

The length of the heat pipe must be estimated based on the dimensions of the system being cooled. However, it cannot be too long otherwise the pressure losses will be too large and the fluid will not circulate. Eq. 2 assumes the mass of the radiator is known, which must be calculated using radiator sizing from Reference [22].



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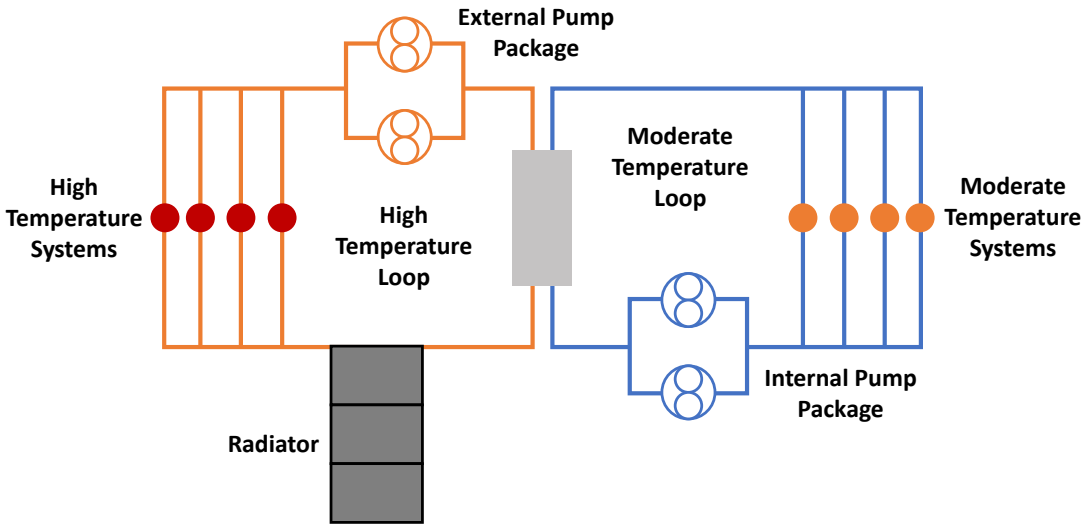



**Fig. 6 Diagram of the active thermal control system [22].**

*3. Habitat Structure*

The habitat is comprised of two parts: a lower rigid section and an upper inflatable section, as seen in Fig. 11. The sizing relations for the inflatable section are described first, followed by the rigid section. The inflatable section consists of many layers: abrasion resistant lining, pressure shell, load-bearing layer, micro-meteoroid and orbital debris (MMOD) shielding, MLI, and beta cloth (ordered from interior to exterior). The abrasion resistant lining protects the more delicate pressure bladder from scratches or punctures. The pressure shell is responsible for containing the gasses that comprise the atmosphere, and usually at least two are included for redundancy. The load-bearing layer is made from woven kevlar so it is strong enough to withstand the loads caused by the internal pressure. The areal density of the first three layers is found using an empirical relationship assuming a cylindrical shape (Eq. 3) or assuming an ellipsoidal dome (Eq. 4)[9].

$$\rho'_c = 1.754 \times 10^{-6} + [0.513 + 0.308(\#_{bladders} - 1)] \cdot p \cdot SF \cdot R_c \tag{3}$$

$$\rho'_d = 1.754 \times 10^{-6} + 1.169 \times 10^{-6} \cdot p \cdot SF \cdot R_c^2 / H_d \tag{4}$$

Areal densities ($kg/m^2$) are denoted with $\rho'$ and typical densities ($kg/m^3$) are denoted with just $\rho$. A safety factor of 1.5 is used in all structure sizing equations. The MMOD layer protects the habitat from being punctured by high-velocity debris, and the MLI layer provides passive thermal control. The exterior layer of beta cloth is resistant to lunar dust and provides some of the thermal characteristics of the inflatable shell. The areal densities of the MMOD shielding and beta cloth are 22 $\frac{kg}{m^2}$ [24] and 0.271 $\frac{kg}{m^2}$ [25], respectively. The MLI areal density depends on the number of layers, because the individual layers of mylar and dacron are added together to get the total density, shown in Eq. 5.

$$\rho'_{MLI} = 0.017\#_{layers} + 0.0063(\#_{layers} + 1) \tag{5}$$

The mass of the cylinder was found with Eq. 6 by multiplying the sum of the areal densities by the surface area, which is trivial to find for a cylinder.

$$m_c = 2\pi R_c H_c(\rho'_c + \rho'_{MMOD} + \rho'_{BetaCloth} + \rho'_{MLI}) \tag{6}$$

The dome type being modeled uses both inflatable and rigid sections, where the inflatable section is a quarter arc that goes around a central flat plate of aluminum that creates the top of the dome. The maximum stress is caused by a combination of the bending loads due to the plate bulging from the internal pressure and the hoop stress. Setting these equal to the yield stress of aluminum results in a quadratic equation (Eq. 7-8).

$$M = p(3 + \nu)(R_c - H_d)^2/16 \tag{7}$$



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$$0 = \sigma_y t_p^2 - p \cdot SF \cdot (R_c - H_d) \cdot t_p - 6 \cdot SF \cdot M \tag{8}$$

Solving the quadratic yields two thicknesses, one positive and one negative. The positive thickness is obviously the physical answer, and can be found using Eq. 9.

$$t_p = \frac{p \cdot SF \cdot (R_c - H_d) + \sqrt{p^2 SF^2 (R_c - H_d)^2 + 24\sigma_y \cdot SF \cdot M}}{2\sigma_y} \tag{9}$$

There is a minimum manufacturing thickness for every material, and for aluminum this was set as 1.02 mm [9]. If any equation resulted in a thickness less than the minimum thickness, it was increased to the minimum value. The surface area of the inflatable part of the dome was found by rotating a quarter circle around the y-axis, shown in Eq. 10.

$$A_i = 2\pi[H_d^2 + \pi H_d (R_c - H_d)/2] \tag{10}$$

The total mass of the dome is the sum of the plate mass, quarter-circle mass, and MLI, beta cloth, and MMOD shielding covering both. The plate mass was found by multiplying the density of aluminum by the volume of the plate. The inner-layer mass of the inflatable part of the dome is found by multiplying the areal density by the area found in Eq. 10. It was assumed that the areal density from Eq. 4 can be used, even though the inflatable shape is not elliptical. The MLI, MMOD, and beta cloth mass was found by multiplying the areal densities by the total surface area of the plate and inflatable curve sections. The equation to calculate the total mass per dome is given by Eq. 11.

$$m_d = \pi (R_c - H_d)^2 t_p \cdot \rho_{Al} + A_i \cdot \rho'_d + [\pi (R_c - H_d)^2 + A_i](\rho'_{MMOD} + \rho'_{BetaCloth} + \rho'_{MLI}) \tag{11}$$

Next the rigid section of the habitat will be sized. The small angled section shown in Fig. 11 was not accounted for, because the angle and length were unknown. Therefore the rigid section is just a cylinder with flat end plates. The thickness of the aluminum in the cylindrical section is found with the equation for hoop stress set equal to the yield stress for aluminum. It can be found using Eq. 12 [9].

$$t_c = \frac{p \cdot SF \cdot R_c}{\sigma_y} \tag{12}$$

The rigid section uses aluminum as the inner layer, but it still has MLI, beta cloth, and MMOD shielding outside of the aluminum to protect the astronauts. The mass of the cylinder can be found using Eq. 13.

$$m_c = \pi H_c [R_c^2 - (R_c - t_c)^2]\rho_{Al} + 2\pi R_c H_c \rho'_c \tag{13}$$

The thickness of the aluminum end plates was found using Eq. 7 and Eq. 9, with $H_d$ set to zero because the dome is flat. The total mass per dome (flat plate) is the sum of the aluminum and outer layers' masses, shown in Eq. 14.

$$m_d = \pi t_p (R_c - t_c)^2 \rho_{Al} + \pi (R_c - t_c)^2 (\rho'_{MMOD} + \rho'_{BetaCloth} + \rho'_{MLI}) \tag{14}$$

Now that the main pressure vessel structure of the rigid and inflatable habitats have been found, other supporting structures must be added to both of them. The dome to cylinder ring interfaces are assumed to be made from composites, and their mass can be found with Eq. 15 [9].

$$m_{ring} = 1.798 \times 10^{-4} \cdot p \cdot SF \cdot R_c \tag{15}$$

The mass of the internal secondary support structure was assumed to have an areal density of 2.46 $kg/m^2$. The mass of the secondary supports is found in Eq. 16 by multiplying by the sum of the dome and cylinder areas.

$$m_{ss} = 2.46(A_c + A_d) \tag{16}$$

The mass of outfitting was assumed to be 10% of the total habitat mass (Eq. 17). This is based on the outfitting percentage of a study where subsystem experts created estimates for the master equipment list of a similar habitat [26].

$$m_{out} = 0.1 m_{tot} \tag{17}$$

The rigid section of the habitat is assumed to have one airlock, which has a mass of 600 kg [26]. Using historical data, the mass of the ground support system (legs), launch integration structure, and non-optimal structures are sized as a percentage of the total habitat mass in Eq. 18-20 [9].



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$$m_{legs} = 0.026 m_{tot} \tag{18}$$

$$m_{int} = 0.02 m_{tot} \tag{19}$$

$$m_{non-opt} = 0.2 m_{tot} \tag{20}$$

The inflatable structure is two stories, so it was assumed to have a floor made from honeycomb composites. The floor panels and supporting structures were sized following the steps described in Reference [9].

*4. Power Generation*

The two methods to generate power that are being modeled are a photovoltaic array and a fission reactor. The photovoltaics system is lighter than the fission reactor, but it requires large batteries to store energy during the lunar night because it cannot generate energy in darkness. The photovoltaics model is for ground-based solar panels with sun trackers to keep the panels oriented perpendicular to the incoming radiation. The solar panels have an operational lifetime of 10 years, and a beginning-of-life efficiency of 17.5%. The yearly degradation was assumed to be 3.75%, because they are silicon cells. Using this information, the end-of-life efficiency of the solar cells can be calculated with Eq. 21 [27].

$$\eta_{eol} = \eta_{bol}(1 - \eta_{deg})^t \tag{21}$$

The solar panels must provide the required power for the base throughout the operational lifetime, so they are sized with the end-of-life efficiency. Using the average solar flux of 1368 $W/m^2$, the power generation per square meter of solar cells can be found using Eq. 22[28].

$$p_{eol} = 1368 \eta_{eol} \eta_{inherent} \cos(\theta) \tag{22}$$

Because sun trackers are being used, the incidence angle should always be near 90° so the cosine term is approximated as unity. In addition to the yearly degradation, there is also an inherent degradation term from manufacturing, which is assumed to be 0.77 [28]. The solar array must provide power for the normal operation of the lunar systems, as well as power to recharge the battery in between periods of darkness. At the South Pole, the darkness is caused by shadows from surrounding terrain features, so the shadow time and frequency is highly dependent on the location. The time in darkness was assumed to last for 100 hours, as per the typical survive-the-night duration [12]. The time in sunlight was taken to be a conservative value of two weeks. The power generation required from the solar array is calculated in Eq. 23.

$$P_{req} = P + \frac{E_{req}}{t_{ssun} - t_{dark}} \tag{23}$$

The total array area can then be found by dividing the power required by the power generation per square meter, shown in Eq. 24.

$$A_{PV} = P_{req}/p_{eol} \tag{24}$$

The solar panels are assumed to have an areal density of 5 $kg/m^2$[27], which can be used to find the PV array mass with Eq. 25.

$$m_{PV} = A_{PV} \rho'_{PV} \tag{25}$$

A fission reactor is another method of generating power. The reactor investigated here uses high-assay low-enriched uranium and produces 40 kW. This produces less power per kilogram, but it avoids problems associated with a highly enriched uranium system. It operates continuously, so there is no need to have more energy storage during the lunar night. The fission reactor is located 1 km from the habitat and any areas the crew may visit, to ensure that the radiation levels are low enough while only using a light radiation shield. The fission reaction is only 18.1% efficient, so a significant amount of waste heat is generated. For this reason the reactor has a built-in thermal control system to maintain component temperatures. The control system for the reactor is located a short distance away, and it also has an individual thermal control system. The total mass of the FSPS is 8934.6 kg and it generates 43.5 kW [16]. The reactor is not sized, instead a single one will be used and if there is additional power generation required it will be handled by a solar array. The power distribution from the reactor is handled by a separate power cable model in Section III.A.6 to calculate the mass and power losses.



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*5. Energy Storage*

Energy storage is required for a lunar architecture in order to power subsystems during the lunar night, and to provide backup energy if one of the power generation sources experiences failure. The two most common energy storage methods for lunar surface systems are regenerative fuel cells and lithium batteries. RFCs are more mass efficient than lithium batteries, but they also produce significantly more heat that must be dissipated by a thermal control system.

Regenerative fuel cells react gaseous hydrogen and oxygen to produce energy, and use electrolysis to split water molecules into hydrogen and oxygen to store energy. There are three aspects to be sized: the storage tanks, which determine the total energy storage, the fuel cell, which determines the power generation capability, and the electrolyzer, which controls the recharge rate. The scaling factors are shown in Table 2.

**Table 2 Scaling factors for the fuel cell (left) and electrolyzer (right)[13].**

| Component | kg/kW |
|---|---|
| Fuel cell stack | 2.00 |
| Hydrogen separation tank | 0.20 |
| Filters | 0.12 |
| Propellant lines and fittings | 0.52 |
| Controller unit | 0.16 |
| Wiring | 0.30 |
| Heat exchanger | 1.00 |
| Water pump | 0.27 |
| Check valves | 0.08 |
| Flow regulators | 0.62 |
| Control valves | 0.16 |
| Pressure and temperature sensors | 0.07 |
| Flow sensors | 0.06 |
| Hydrogen phase separator | 0.07 |
| Hydrogen regulator | 0.05 |
| Oxygen separation tank | 0.18 |
| Oxygen regulators | 0.09 |

| Component | kg/kW |
|---|---|
| Electrolyzer stack | 2.00 |
| Water tank and heater | 0.10 |
| Filters | 0.12 |
| Propellant lines and fittings | 0.52 |
| Controller unit | 0.16 |
| Wiring | 0.30 |
| Heat exchanger | 1.00 |
| Water pump | 0.27 |
| Check valves | 0.08 |
| Flow regulators | 0.62 |
| Control valves | 0.16 |
| Pressure and temperature sensors | 0.07 |
| Flow sensors | 0.06 |

The total mass of the stored hydrogen can be calculated using the required energy storage. The fuel cells must store enough energy to provide power during the 100 hour survive-the-night scenario. It is assumed that the power load during the night is 50% of the power load during the day, because nonessential systems are shut off and other systems are operated at a lower power to conserve energy. The energy storage required was calculated with Eq. 26.

$$E_{req} = \frac{0.5Pt_d}{\eta_{fc}} \tag{26}$$

The efficiency of the regenerative fuel cells was set at 62.2% [12]. The reaction is stoichiometric, consuming one mole of oxygen for every two moles of hydrogen. Oxygen is 16 times heavier than hydrogen, so the total mass of the stored oxygen is 8 times that of the hydrogen[13]. The mass of the hydrogen and oxygen that need to be stored were calculated using Eq. 27-28.

$$m_{H2} = \frac{E_{req}}{39273\eta_{fc}} \tag{27}$$

$$m_{O2} = 8m_{H2} \tag{28}$$

To make the storage simple and cost-efficient, the hydrogen and oxygen are stored in off-the-shelf composite overwrapped pressure vessels from Northrop Grumman. The number of pressure vessels was found by dividing the total mass of the hydrogen or oxygen by the storage mass per container, and rounding up to the nearest integer.



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The other energy storage option being modeled is lithium-ion batteries. The batteries must provide the same amount of energy, but they have a higher efficiency of 90% so the storage capacity is lower than the regenerative fuel cells [27]. The energy storage required for the battery was calculated using Eq. 29.

$$E_{req} = \frac{0.5P \cdot t_d}{D_d \eta_{bat}} \tag{29}$$

The rate of degradation for the battery depends on the discharge depth. The battery will not go through many charge-discharge cycles because the lunar night does not occur frequently, so the discharge depth was set relatively high at 80%. A high discharge depth reduces the mass of the battery because a higher percentage of the stored energy can be used, so the required capacity of the battery is reduced. The mass of the battery depends on the specific energy, which for a lithium ion battery is 125 $Wh/kg$ [28]. With the energy required and the specific energy, the battery mass can be calculated with Eq. 30.

$$m_{bat} = E_{req}/E_{req} \tag{30}$$

*6. Power Distribution System*

Power transmission is important to consider because many of the lunar surface systems will not be located close to each other. The power generation sources may be located away from the base: in the case of the fission reactor it is necessary to reduce radiation exposure for the crew, and the photovoltaic system may be at a nearby raised spot to increase sunlight exposure. The ISRU system requires regolith with higher concentrations of ilmentite, which will not necessarily occur in the vicinity of the habitat or power generation system. A previous study investigated sizing three different power distribution systems: DC power cables, radio frequency power beaming, and optical power beaming. Transmission distances of 1 to 15 km and power levels of 10 to 50 kW were sized. For each system, every component was individually sized as a function of the distance and power. Due to inefficiencies, some of the power is converted to heat at different steps along the transmission process. An individual active thermal control system was sized for each stage that required cooling, and this is included in the final mass estimations. After running many different sizing scenarios, the resulting power distribution system masses were compared. Fig. 7 shows the design space and where each system dominated by having the lowest mass[29].

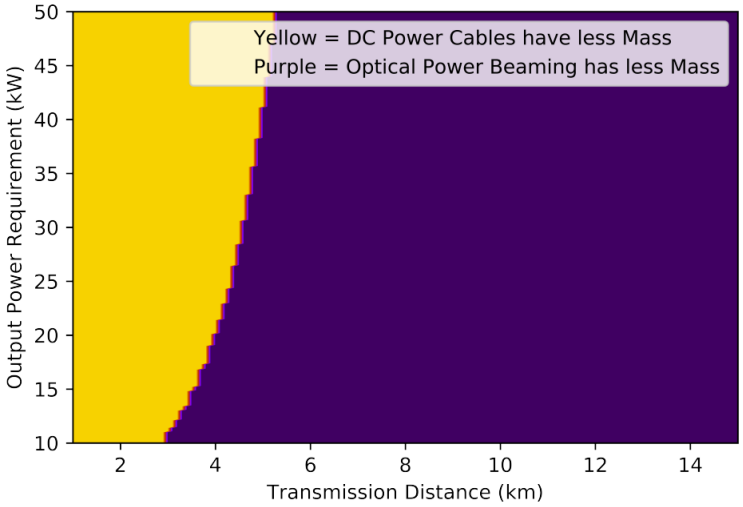



**Fig. 7 Power distribution system with the least mass[29].**

The radio frequency power beaming system is never the lightest, so it will not be used. For more advanced lunar architectures, the base may be more spread out and the power requirements will be higher, so an optical power beaming would be the mass efficient choice. However, the lunar base architectures considered here are variations of an initial lunar base, so there are only a few systems and they are located within 3 km or less of each other. Therefore only DC power cables will be used, as they are the lightest for the cases being studied.



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The power cable can be buried beneath the surface of the regolith, resting on the surface, or suspended above the regolith like transmission lines on Earth. The cable was selected to be resting on the surface, because the deployment is simple and doesn't require a burying system or poles to support the wires. To account for uneven terrain and avoiding obstacles, the length of the cable is assumed to be 10% longer than the distance between the two objects it connects[30]. The cable is coaxial, so the two conductors are separated by a layer of dielectric insulation to prevent the cable from short-circuiting. The conductor was chosen to be aluminum because it is much less dense than copper, and the dielectric selected was TFE fluorocarbon. The properties used for these materials when sizing the power cables are listed in Table 3.

**Table 3 Properties of aluminum and TFE[30].**

| | Density $(kg/m^3)$ | Thermal Conductivity $(W/m-K)$ | Base Resistivity $(\Omega - m)$ | Breakdown Strength $(V/mil)$ |
|---|---|---|---|---|
| Aluminum | 2700 | 237 | $2.83 \times 10^{-8}$ | - |
| TFE | 2175 | 0.251 | - | 430 |

The cables are sized to transfer a specific amount of power, so they should be sized after the other systems once the power load is known. The first step of the sizing process is to assume a voltage. In this study a list of voltages were used, and the value that produced the smallest mass was selected. Using the voltage, the current and dielectric thickness can be calculated with Eq. 31 and Eq. 32, respectively.

$$I = P/V \tag{31}$$

$$t_d = V/V_b \cdot 10 \tag{32}$$

A safety factor of 10 is applied to the dielectric thickness, to account for any degradation of damage after deploying the wires. The wire is assumed to have a 95% transmission efficiency. A higher transmission efficiency will increase the mass of the wire because the conductor will be larger, and a lower transmission efficiency will increase the required power generation, so the value should be selected with that trade-off in mind. The efficiency of the line determines how much of the power is transformed into heat (Eq. 33), and power lost is directly related to the resistance of the line, shown in Eq. 34.

$$P_{lost} = 0.5P(1 - \eta_w) \tag{33}$$

$$R = P_{lost}/I^2 \tag{34}$$

$P_{lost}$ is not the total power lost, rather the power lost per line. Both lines have the same properties, so they are each responsible for half of the power losses. The resistivity of the conductor is a function of temperature, and it is assumed to scale linearly over the range of temperatures being used. Starting with the base resistivity given in Table 3, the temperature the base resistivity occurs at (293 K), and an initial temperature guess of 500 K, the resistivity can be found using the linear relationship with Eq. 35 [30].

$$\rho_w = \rho_0 \frac{T_w - 45}{293 - 45} \tag{35}$$

The resistivity and the resistance of the wire are related through the cross sectional area of the wire and the total length. This relationship can be used to solve for the area of the wire, which is taken one step further to find the radius of the wire in Eq. 36 [30].

$$r_i = \sqrt{\frac{\rho_w L}{\pi R}} \tag{36}$$

The above equation finds the radius of the inner conductor. The outer conductor is a hollow cylinder, containing both the dielectric and the inner conductor inside of the hollow area. The inner radius of the outer conductor is the sum of the two radii it encloses. The naming convention of the radii is shown in Figure 8.

Both conductors must have the same cross sectional area in order to have the same resistance, which can be leveraged to solve for the outer radius of the outer conductor. The radius of the dielectric was calculated with Eq. 37 and the outer radius of the outer conductor was calculated with Eq. 38.



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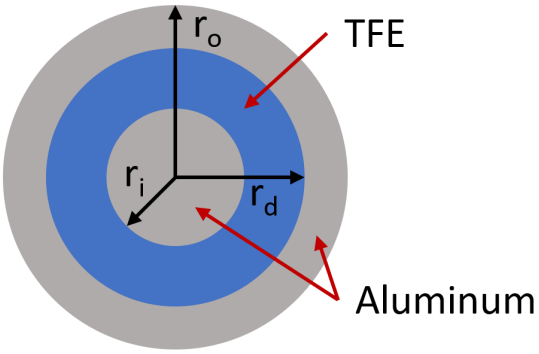



**Fig. 8 Cross-sectional view of the coaxial cable laying on the regolith.**

$$r_d = r_i + t_d \tag{37}$$

$$r_o = \sqrt{r_i^2 + r_d^2} \tag{38}$$

There are two sources of heat that affect the wire: internal generation from resistance and incident radiation from the Sun. The internal generation is assumed to be uniform throughout the volume of the conductor. In this paper the lunar base is located at the South Pole, so the radiation is approximately parallel to the ground. The cable is assumed to be half-buried by the regolith, so the projected area that the radiation will hit is a rectangle the length of the wire with a height equal to the outer radius. A diagram of these heat sources is shown in Fig. 9.

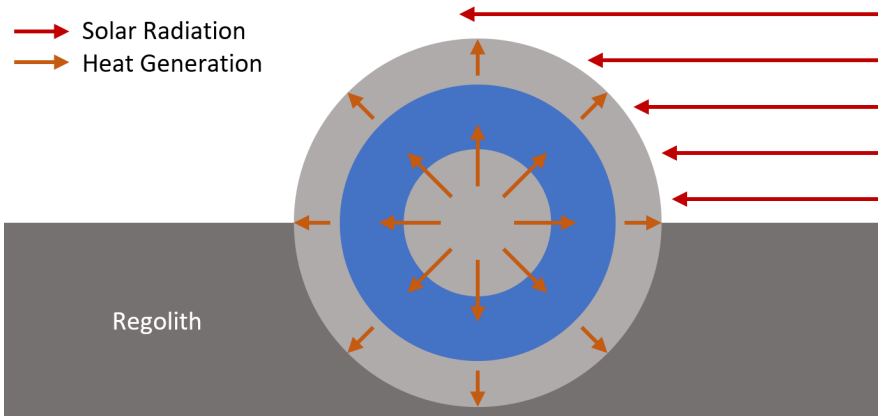



**Fig. 9 Incident radiation and internal generation of the cable.**

The heat generated is the same for both wires, because they have the same cross-sectional area and power losses. Eq. 39 can be used to calculate the rate of heat generation. The amount of radiation absorbed by the wire is determined by the absorptance of the outer material, in this case aluminum, and can be calculated using Eq. 40.

$$q_{gen} = P_{lost}/(\pi r_i^2 L) \tag{39}$$

$$q_{rad} = 2\alpha r_o L q_{in} \tag{40}$$

The incident radiation value used was 1421 $W/m^2$, which is the maximum value that occurs when using the solar constant and the minimum Moon-Sun distance [31].

There are two sources of heat dissipation: the upper-half of the wire radiates heat to the surrounding regolith and space, and the bottom-half conducts heat to the regolith. Because of this, the area value used in each equation is half of the total surface area, demonstrated in Eq. 41.



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$$A = \pi r_o L \tag{41}$$

The wire is in thermal equilibrium, so the temperature of the wire remains constant and the dissipation is equal to the generation. The outer temperature is solved first to ensure the wire is in equilibrium with the surroundings. Assuming the cable is a half cylinder radiating to an infinite plane, the view factor is 0.25. This assumes the plane is flat and the cable is perfectly straight so it cannot radiate to itself. Therefore 25% radiates to the regolith at 224 K [31], and the remaining 75% of the top surface radiates to space at a temperature of 2.7 K [32]. Eq. 42 can be numerically solved to find the outer temperature of the wire.

$$2P_{lost} + q_{rad} = \epsilon\sigma_{sb}A\left[F(T_o^4 - T_s^4) + (1-F)(T_o^4 - T_{space}^4)\right] + \frac{A(T_o - T_s)}{R_{reg}} \tag{42}$$

With the outer temperature known, the temperature at $r_d$ can be found. The first step is to integrate the heat conduction differential equation in cylindrical coordinates with an internal generation term, and solve for the integration constants using the $T_o$ and $T_d$ as boundary conditions. Then take the derivative and set equal to the power loss of the line using Fourier's law for heat conduction. The result is Eq. 43, after rearranging to solve for $T_d$ [30].

$$T_d = \left(\frac{P_{lost}}{2\pi k_w L} - \frac{q_{gen} r_o^2}{2k_w}\right)\ln\left(\frac{r_o}{r_d}\right) + \frac{q_{gen}(r_o^2 - r_d^2)}{4k_w} + T_o \tag{43}$$

Performing the same steps, but this time without the internal generation term and with $T_i$ and $T_d$ as boundary conditions, results in an Eq. 44 [30]. This is used to find the temperature increase across the dielectric layer to determine the inner wire temperature.

$$T_i = \frac{P_{lost}\ln(r_d/r_i)}{2\pi k_d L} + T_d \tag{44}$$

The highest temperature in the cable is $T_i$, which must be below the maximum operating temperature of 573 K [30]. Any voltage values that produced a temperature above the maximum temperature were excluded from being selected as the final voltage. With the wire temperature known, the initial guess in Eq. 35 should be replaced with the calculated value, and the process should be repeated until the calculated temperature has converged. With the dimensions of the cable converged, the mass can be found using Eq. 45.

$$m_{cable} = 2\pi r_i^2 L\rho_{Al} + \pi(r_d^2 - r_i^2)L\rho_d + \rho_{emi}L \tag{45}$$

The inner and outer conductors have the same volume, so the inner one is doubled and added to the mass from the dielectric layer. The extra term is included to account for electromagnetic interference shielding mass. The value used in this study was 0.15 kg/m, based on Reference [33] assuming some mass reduction improvements. This was repeated for each voltage, and the voltage that resulted in the lowest mass was chosen. The dimensions associated with the final voltage were checked against ASTM standards to ensure that the wire thickness could handle the required amount of current.

The cable must be stored in a spool for transportation and deployment. A NASA study tested cable deployment in a desert, and the spool dimensions were estimated from technical drawings of the deployment system to come up with Eq. 46 to calculate the storable length[34].

$$L_{store} = \frac{10.5(41.5^2 - 21.875^2)}{3.281 \cdot 30.6 r_o^2 \cdot 39.3701^2} \tag{46}$$

The total number of spools was found by dividing the cable length by the storable length, and rounding up to the nearest whole number. The mass of an individual spool is 10 kg[34].

In order to step up the voltage from the power generation source to the transmission voltage chosen above, a DC to DC conversion unit (DDCU) is required. Likewise, at the end of the power cable another DDCU is necessary to step down the voltage from the transmission voltage to 120 V. The DDCUs were sized following the process detailed in Reference [35]. Each DDCU must have a thermal control system, because they have inefficiencies that generate heat when changing the voltage. The DDCUs use the heat pipe and radiator models from Section III.A.2.



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*7. In-situ Resource Utilization*

In-situ resource utilization makes a lunar architecture more sustainable by using local resources to create products. ISRU can produce water or oxygen which can replace what is lost by the ECLSS, removing the need to resupply water or oxygen. The Moon is also being used as a test bed: ISRU technologies will be demonstrated and refined on the lunar surface before being used for Mars missions. The type of ISRU being investigated involves reacting hydrogen with ilmenite to produce water, which is shown in Eq. 47. Ilmenite is an iron titanium oxide and can be found at almost any location. Surface regolith with high iron and titanium concentrations can be used for ISRU, removing the need for any complex and power intensive mining equipment. After the water is extracted, the hydrogen and oxygen are separated with electrolysis; the hydrogen can then be reused for the reaction and the produced oxygen is stored[13].

$$FeTiO_3 + H_2 \rightarrow Fe + TiO_2 + H_2O \tag{47}$$

The ISRU is composed of an ilmenite reduction reactor, electrolyzer, extraction rovers, and an active thermal control system. The thermal control system is comprised of a single loop with ammonia as the working fluid and a double-sided radiator to remove heat. It is sized in the same manner as the active thermal control for the habitat. The ISRU system is sized to generate oxygen at a rate equal to the crew consumption. The water load required by the electrolyzer is calculated from the oxygen load using the molar masses in Eq. 48.

$$\dot{m}_{h2o} = \frac{18}{32}\dot{m}_{o2} \tag{48}$$

The water mass flow rate has the same units as the oxygen flow rate (kg/day). The change in Gibbs free energy, which is dependent on the temperature of the water, is calculated in Eq. 49[13]. It is assumed the water enters the electrolyzer at 340 K. The power consumed by the electrolyzer is calculated with the Gibbs free energy in Eq. 50 [13].

$$\Delta G = (285.83 \times 10^3 - T_{h2o}163.295)/18 \tag{49}$$

$$P_e = \frac{\Delta G \dot{m}_{h2o}}{24 \cdot 60 \cdot 60 \cdot \eta_e} \tag{50}$$

The change in enthalpy of water is assumed to be a constant value ($285.83 \times 10^3$ J/mol) and the change in entropy is $163.295\ J/mol - K$. The efficiency of the electrolyzer is 72.7% [13]. The electrolyzer mass is calculated using the same relationships as the regenerative fuel cell in Section III.A.5.

The extraction rover has the capacity to move 114.15 kg of regolith per hour, assuming it operates for 16 hours and recharges for 8 hours with a consistent 24-hour cycle. The number of rovers needed is calculated using Eq. 51 [36].

$$\#_{rovers} = ceil(\dot{m}_r/114.15) \tag{51}$$

The number of rovers to meet the load is rounded up to the nearest integer, so the rovers can work below full capacity to meet the regolith load requirement. One backup rover was also included in the sizing, so if any rover fails there is a replacement. Each rover weighs 66 kg. The power used by the rovers depends on the required regolith load, and can be calculated with Eq. 52 [36].

$$P_{rover} = \dot{m}_r 4 \cdot 16/8 \tag{52}$$

The specific power of the rover per regolith flow rate is 4 $\frac{W-hr}{kg}$ [36], so when multiplied by the regolith flow rate and operation duration of 16 hours, it results in the energy consumed by the rover. This energy must be recharged over the 8 hour recharge duration, resulting in the power consumption of the rover. The ilmenite reduction reactor is sized following the process described in Reference [13]. It is sized to generate oxygen equal to the crew consumption rate. The reactor also produces a small surplus of hydrogen, which is stored and eventually sent to the carbon-dioxide reduction system in the ECLSS to produce additional water.

*8. Mobility*

Mobility is necessary to explore the lunar surface and conduct science experiments at different locations. The astronauts also must be able to transport, deploy, and fix the different systems. A pressurized lunar rover allows the astronauts to travel farther from the lunar base and stay away for days or weeks at a time. This is an enabling technology for many science and exploration missions. The pressurized lunar rover is sized using the tool created in Reference [37].



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The tool also was created in OpenMDAO, and works by sizing the subsystems of the rover: ECLSS, thermal control, power generation, energy storage, movement, and structures. The subsystems are sized using mission parameters, like the number of crew members, and design choices like the number of wheels.

For shorter duration missions, such as a system repair or science near the base, an unpressurized lunar rover is ideal. The astronauts remain in their space suits, so they do not have to go through the process of entering and exiting the pressurized rover and spacesuits each time they need to perform a task. Using an unpressurized rover in place of a pressurized one can save time and resources for some missions. The unpressurized rover is being adapted from the pressurized rover sizing tool, by removing unnecessary elements like the ECLSS and pressure vessel from the sizing process.

### B. Baseline Sizing

The Artemis Planning Manifest by the NASA Advisory Council outlines in detail the steps that must be accomplished for Artemis I through Artemis IV. It also includes a tentative timeline for when different lunar systems will be operational, as well as when smaller satellite and rover exploration missions will occur. The baseline case was selected from the manifest; it outlines a plan to have a lunar base with the necessary power generation, an unpressurized lunar rover, and a pressurized lunar rover on the Moon by 2031[3]. The surface habitat will be near the Shackleton Crater rim, and will have an initial capacity to house two crew for 30 days, with the ability to expand and eventually hold four crew for up to 60 days[6, 38]. Therefore the baseline habitat will be sized for two crew and a 30 day mission, with 7 contingency days. The ECLSS system baseline uses the regenerative technologies present aboard the ISS, including an oxygen generation system, water recovery system, carbon-dioxide reduction system, and waste processing system. The habitat is comprised of two sections: the lower one is a rigid single story and houses the airlock, and the upper one has two floors and uses inflatable technology. The structure shape and baseline dimensions are shown in Fig. 10.

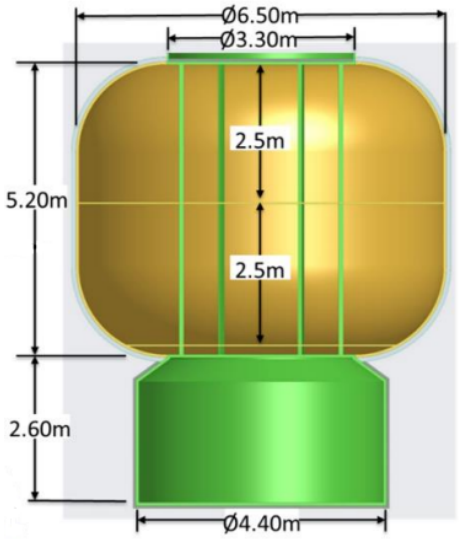



**Fig. 10 Dimensions of the baseline habitat[39].**

Photovoltaics will be used to generate power because they are the standard for crewed missions[12]. It is ideal for the photovoltaic system to be located at the highest possible local elevation in order to prevent shadows from reducing the efficiency. For the baseline it was assumed that the solar panels are placed 1 km from the base, and the power is transferred via DC cables resting on the lunar surface. During the survive-the- night scenario of 100 hours of darkness, the power to systems is provided by a battery pack located near the habitat. During the night non-essential systems are turned off, so the power requirement was assumed to be half of the operating daytime value. In NASA's unpressurized rover industry Request for Information, one requirement was that the unpressurized rover must be able to carry two crew at least 2 km on a single charge[40]. At a cruise speed of 15 kmh, this distance is achieved in less than fifteen minutes. To compare, Lockheed Martin's proposed design has a range of 300 km per charge [41]. A more reasonable baseline was chosen to be a 50 km range, which provides a little under eight hours of drive time per charge when operating at cruising speed. The unpressurized rover does not have a power generation system, so it must recharge by plugging



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into the electrical network of the habitat. The baseline pressurized rover must support two crew for a duration of 14 days[42, 43]. The pressurized rover has solar panels so it can recharge during long missions, therefore it was assumed to not draw any power from the lunar grid, which is why it is separate from the other systems in the diagram of Fig. 11.



**Fig. 11 Diagram of the baseline lunar architecture[5–7].**

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### C. Variation from Baseline

To investigate how design decisions affect the lunar architecture, different aspects of the baseline were changed to create new architectures. These new designs were then compared against the baseline using common figures of merit like the gross mass and power consumption. The first variation involves replacing the oxygen generation of the habitat with an ISRU system, shown in Fig. 12. The ilmenite reduction system is sized to produce oxygen equal to the rate of consumption by the crew, and the oxygen is stored in a pressurized tank at the ISRU plant. The tank is sized to hold two weeks worth of oxygen, so every other week the crew must drive to the ISRU plant to exchange the full and empty tanks. The ISRU system also captures a small amount of hydrogen that was trapped in the regolith, which is stored in a separate tank. This hydrogen can be retrieved and fed into the carbon-dioxide reduction system to produce a small amount of additional water. The ISRU system is most effective in a location with a higher concentration of ilmenite, and it does not need to be located near the habitat. Because of this, the ISRU system has a separate solar array and battery system to power the production and survive the lunar night, independent of the habitat grid. The ISRU system must have a battery because continuous production is assumed, so it must function in the dark.

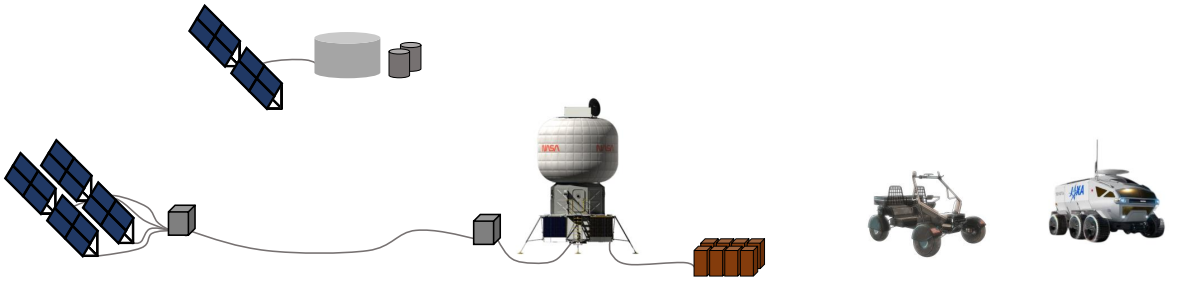


**Fig. 12 Diagram of the lunar architecture with added ISRU[5–7].**

The second variation replaces the photovoltaic power generation system with a fission system. The FSPS is assumed to be the same distance away from the habitat (1 km) and transmit power through the same DC power cables. The FSPS can generate up to 40 kW, which is much higher than the power consumption of the baseline. Another advantage is that the FSPS generates power continuously, so there is no need for a battery during the lunar night. As the lunar base grows and the power demand increases, the FSPS will be able to provide the necessary power so the mass of the power system remains constant. For the PV system, as the base grows the power system mass increases because the array will need to be larger to provide more power, and additional batteries will need to be added to provide power in the darkness.

The third variation builds off the second one by adding additional power consuming systems. The systems are assumed to be independent and not affect the sizing of the baseline systems, so they are just modeled as a power load. The mass of the power generation and distribution systems will be compared to see when the fission system offers a mass savings. The additional systems could be scientific, construction, movement, etc., but are left unspecified so the results will apply to any combination of systems that produce an equivalent additional power load.

The final variation adds an FSPS and replaces the oxygen generation system of the habitat ECLSS with ISRU. The

FSPS now provides power to both the habitat and the ISRU plant, replacing the separate ISRU solar array and battery system that was initially used to generate power. The power is transferred via a second DC power cable, with another set of step-up and step-down DDCUs. The ISRU is assumed to be 3 km away from the FSPS, because it will not be on the steep crater rim and must be near a local deposit where the ilmenite concentration is higher. The base layout for this variation is shown in Figure 13.

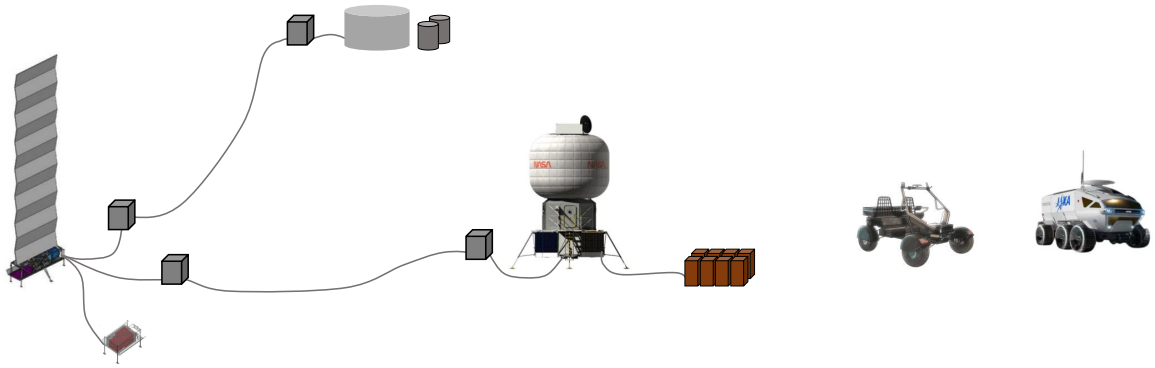


**Fig. 13 Modified baseline layout to include ISRU and fission power generation[5–7, 44].**

A summary of the parameters and their variation from the baseline architecture is shown in Table 4.

**Table 4 Design parameters.**

| Parameter | Baseline | Alternative |
|---|---|---|
| $O_2$ production | ISS OGA | ISRU $H_2$ Reduction System |
| Power generation | PV System | fission surface Power system |
| Habitat crew size | 2 | 4 |
| Habitat mission duration | 30 days | 7-60 days |

The crew size and mission duration were also varied to demonstrate the effects of some design choices that have associated dependencies. The upper limit of four crew and 60 days was chosen because it is the expansion capability requirement for the surface habitat. A reasonable lower limit of one week was chosen, because any shorter mission has little value relative to the cost and difficulty of getting to the lunar surface.

## IV. Results

The baseline case was sized according to the requirements in Section III.B. A high level MEL of all of the systems is shown in Table 5.

**Table 5 Baseline architecture MEL.**

| System | Basic Mass (kg) | MGA (kg) | Predicted Mass (kg) |
|---|---|---|---|
| Habitat | 11820.5 | 518.1 | 12338.6 |
| PV array | 150.0 | 45.0 | 195.0 |
| Battery | 3445.8 | 861.5 | 4307.3 |
| Power cables | 422.7 | 84.5 | 507.2 |
| Pressurized rover | 4997.6 | 1249.4 | 6247.0 |
| Unpressurized rover | 1261.6 | 315.4 | 1577.0 |

Mass growth allowances (MGA) were included at the subsystem level. Some models already included an MGA calculation, but any that did not had a percentage applied following the guidelines of Reference [23]. A margin of



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20% was also applied to the power loads, so the energy storage and power generation systems are sized to account for any additional inefficiencies. The resulting masses match the expected values from literature. The pressurized rover mass varies depending on the size and mission it is designed for, but values between 3,000 and 10,000 kg have been proposed [45, 46]. Following the Artemis Planning Manifest, the surface habitat is required to be under 12 tons. The basic mass of the habitat meets this requirement, but the predicted mass exceeds it by approximately 350 kg. Technology improvements or detailed designs are necessary to reduce the mass and meet the requirement. One possible area of improvement is in inflatable technology because it is the heaviest habitat component. A breakdown of the habitat masses is shown in Fig. 14.

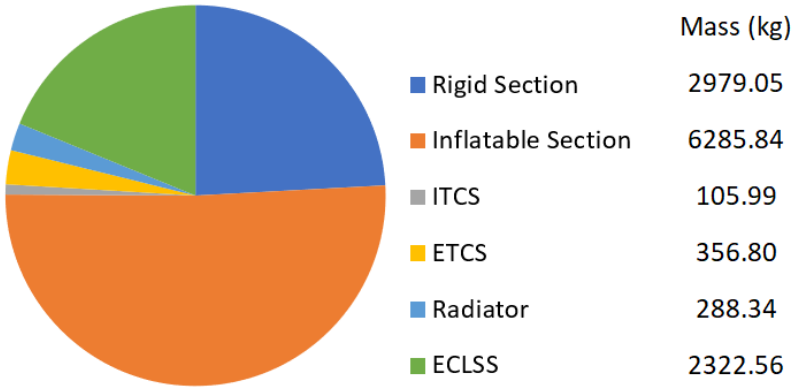



**Fig. 14 Breakdown of habitat subsystem masses.**

The first variation from the baseline is adding an ISRU system to generate oxygen for the crew, which will replace the oxygen generation system in the habitat. The mass will change with mission duration, because the ISRU lowers the required storage for some ECLSS resources by producing them locally. The gross mass of the lunar base is plotted against mission duration with and without ISRU in Fig. 15.

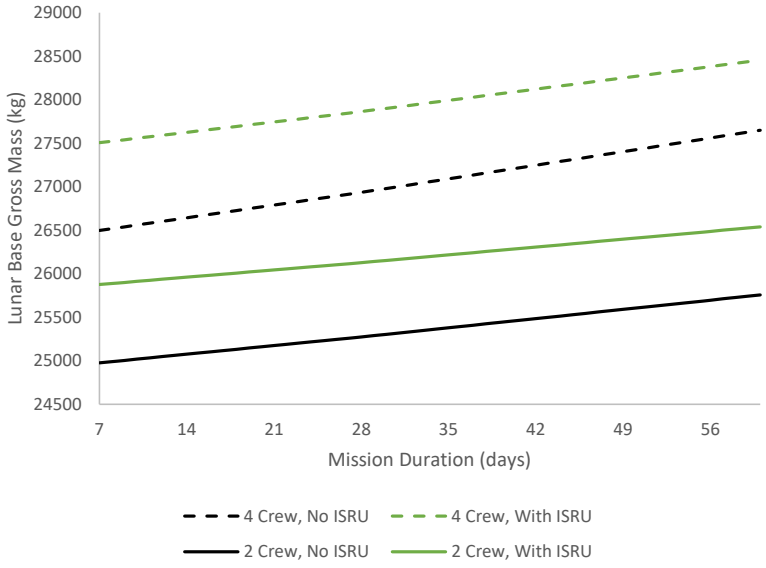



**Fig. 15 Effects of ISRU on the lunar base gross mass with two or four crew.**

The ISRU system has a higher mass than the OGA, so for a single mission the total mass is higher. When the number of crew is increased from two to four, the ECLSS system is larger and must store more food and water, so the total mass increases. The mass increase is about the same with and without ISRU; however, because the oxygen consumption is higher with four crew, the water savings are greater. Therefore, with a larger crew the intersection point will occur



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sooner and the ISRU system will pay off quicker. The percent increase of ECLSS mass is plotted for two and four crew in Fig. 16.

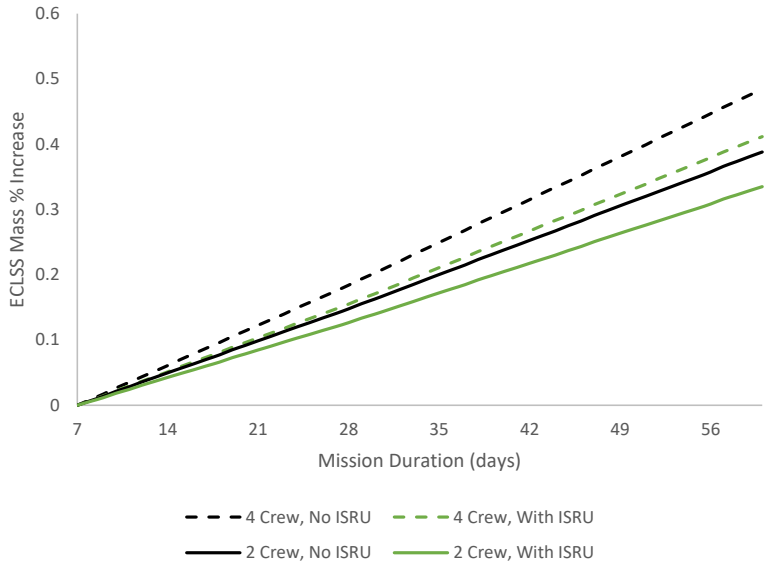



**Fig. 16 Percent increase in ECLSS mass as mission duration increases.**

As the mission duration increases, the ECLSS mass increases slower when there is an ISRU system. There are three main resources that humans consume, which are therefore dependent on the mission duration: food, water, and oxygen. Water is also used to produce oxygen with OGA, thus only food and water are stored. With the ISRU producing oxygen, water is only needed for crew consumption, therefore the storage requirement per day of the mission is lower. If many lunar missions are considered, there is a point where the ISRU can save mass because less water must be delivered and stored on the surface. The intersection point will depend on the number of crew as well as the total time on the lunar surface. With two crew the ISRU system saves 15.6 kg per week, so after approximately one year of occupation the ISRU system will be lighter than the baseline. This is a very rough estimate, as resupply mass is more complex and not calculated the same way as increasing the mission duration. The next variation from the baseline that was investigated used a fission system to replace the solar array, the resulting masses with two and four crew are shown in Fig. 17.

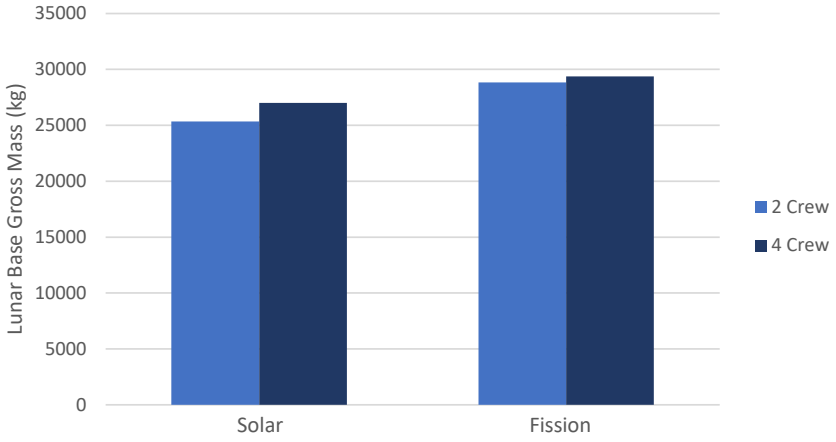



**Fig. 17 Habitat gross mass with solar or fission power.**

The mass of the FSPS is heavier than that of the solar array and battery, so it is heavier than the baseline. However, the FSPS can generate more power than what is required for the baseline, thus it will not increase in mass as the



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architecture grows. If the number of crew is increased to four, the ECLSS is heavier, so the lunar base gross mass grows for both cases. The architecture with the PV generation has a larger mass increase because the PV array and battery size also have to increase to compensate for the higher load. If additional power loads are added to the architecture, the solar array and battery must keep growing, whereas the FSPS will be a constant mass until its power generation capabilities are exceeded. By increasing the power load, the critical power can be found such that the FSPS has a lower mass than the baseline. Fig. 18 plots the mass of both systems to find where the intersection occurs.

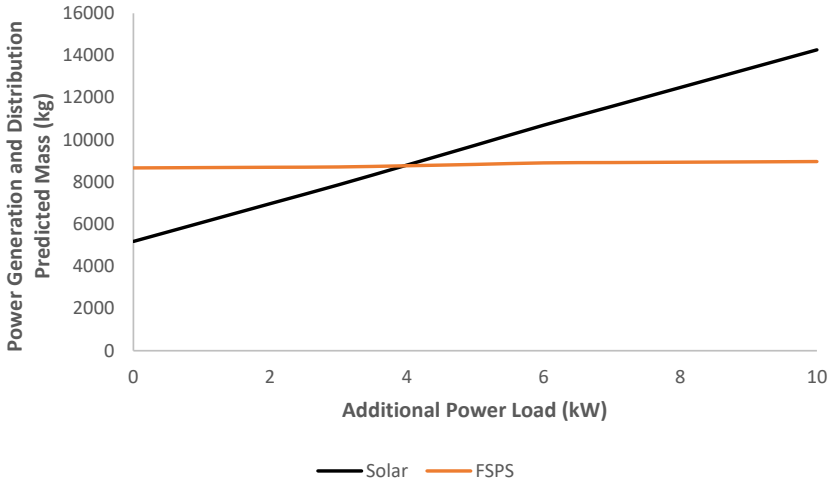



**Fig. 18 Power system mass intersection between fission and solar power**

When the power demand grows 4 kW above the baseline power (~10.2 kW total), the FSPS system is lighter than the solar array and battery. Finally, when both an FSPS and ISRU system are included, the mass is much higher than the baseline and it is higher than each of the individual variations. However, once the base has grown, this is no longer the case. All of the variations were sized for the expanded habitat case with four crew and a 60 day mission duration. The base has grown and there are likely new systems that have been added, therefore they are also sized with additional power values of 3 and 5 kW. Fig. 19 plots the masses for the previously described scenarios.

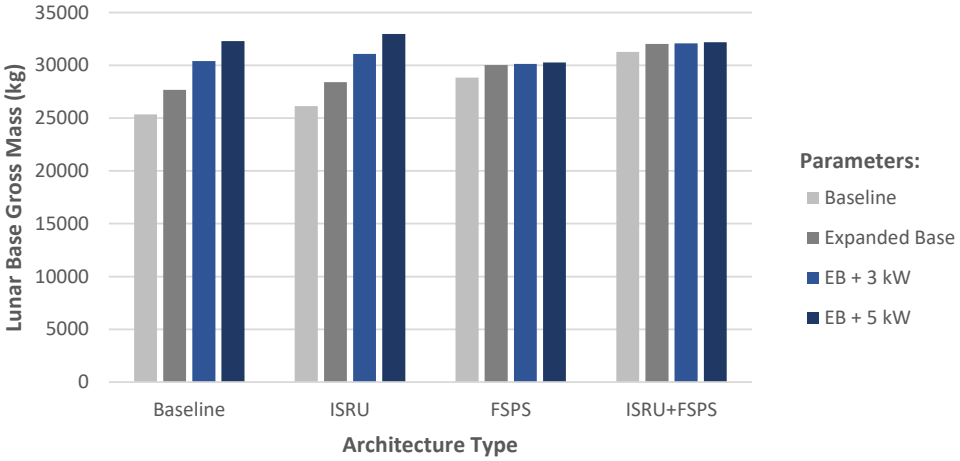



**Fig. 19 Comparison with four crew and a 60 day mission.**

The baseline architecture is still the lightest for the baseline parameters and the expanded habitat parameters. If the expanded habitat has additional systems that increase the power load by 3 kW, the architecture with just the FSPS is now lighter than the baseline and the ISRU architectures. The ISRU plus FSPS architecture is still heavier, because the ISRU saves mass over time and the mission duration of 60 days is still relatively short. Once the additional power climbs to 5



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kW, the ISRU plus FSPS variation is now lighter than the baseline. The variation with only the FSPS is the lightest, but if many missions are performed, the ISRU plus FSPS architecture will pay off in the long run.

## V. Conclusion

Space agencies around the world are working to build a lunar architecture and establish a sustained lunar presence. The Moon is a proving grounds to demonstrate the capability and reliability of new systems, which will then be used to support human life on Mars. The lunar architecture cannot be created all at once, it must be built piecemeal over time. The lunar surface systems must be correctly chosen and sized to enable the desired capabilities. The methodology covered here sizes all lunar systems for a desired architecture together. This ensures the final architecture will be viable, even if the systems are created and delivered to the Moon across a large time span. The results of this study can be used to inform decisions: trade-offs can be compared side-by-side to see the system changes, and the responses of the system to varying mission parameters can be visualized. Short-term, the baseline case is the best option because it has the least mass and all of the systems have a high TRL. Looking ahead, unproven technologies like ISRU and fission power add initial mass and complexity to the architecture, but can save mass in the long run once the base is large enough and the crew have spent substantial time on the lunar surface.

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